Duct Inlet Design Rules

How to shape the inlet of a static ducted propeller so the duct adds thrust instead of costing it: tip gap, lip radius, wall thickness, where the propeller sits, and the outlet. Written for the APC 8x4.5MR duct of the 2026-10-27 research tree. Every number in the rules is someone else's, with the conditions it was measured under.

2026-11-02
derived from EXTRACTS-DUCT-LINING.md
§1–1.4 · 278 quotes verified
quotes below re-checked
against the PDF text dumps
evidence: papers/SoundVisualizer/
design-rules/

IN SHORT Close the blade-tip gap to about 1 mm first; then give the inlet a generously rounded lip (10–13 % of the throat diameter, about 20–26 mm here) on a wall at least 20 mm thick, with the propeller set back behind the lip. The outlet flare is second-order. On a duct this small, getting the gap and the lip wrong turns the duct from a thrust gain into a thrust loss — which is what our unlined duct measures today.

How to read this page. It is a derived document: it summarises `papers/SoundVisualizer/design-rules/EXTRACTS-DUCT-LINING.md` §1–1.4 (private papers repo), and where the two disagree **the extract governs**; where an extract and the paper disagree, the paper governs. Each rule carries its evidence type — **E** experiment, **S** simulation or analytical model, **R** reported second-hand inside another paper — and a file and PDF page. Millimetres are worked out for the 8-inch propeller (D ≈ 203 mm, R ≈ 102 mm) with the throat diameter D_t taken as ≈ D; our duct's real dimensions are not yet recorded, so treat them as starting values.

01 Why it matters here — our rig, measured

2026-11-02, ESC signal 2000 µs, duct and propeller one piece on the Tyto stand (the load cell sees both)

SET-UP	THRUST	RPM	CURRENT
Bare propeller, no duct	−10.57 N	11,804	22.94 A
Duct, no lining	−8.01 N (repeat −7.93 N)	12,262	20.97 A
Duct + Thinsulate 2 / 8 / 10 layers	−8.77 / −8.82 / −8.91 N	12,145–12,212	21.2–21.4 A
Duct + Thinsulate 4 layers	−9.23 N (repeat −9.23 N)	12,140	21.45 A

The empty duct gives 24 % less thrust than the bare propeller while turning 4 % faster; every lining, which narrows the tip gap, wins part of it back. A well-shaped duct should do the opposite of the first line (§03). These are our captures (bases `axi2217__8in__…__r27-*`), not literature; repeat runs of one set-up agree within 0.1 N on the same day.

02 The rules, in order of importance

RULE	NUMBER AND CONDITION	FOR OUR DUCT	EV.	SOURCE
1. Tip gap: as small as you can build, about 1 mm	5-inch Gemfan, CFD: 0.6 % and 1.1 % of R perform alike; 3.6 % R “a sudden drop”. Reducing 3.04→1.71 % R: +17.85 % hover efficiency; 3.04→5.17 % R: up to −18.1 %, below an open rotor. 10-inch duct, measured: a wider gap collapses the suction on the lip and with it the duct thrust. NASA rig: 0.05 in (1.27 mm), limited by misalignment and blade stretch.	aim ≤ 1–1.5 mm; 3.7 mm already bad; 5.3 mm worse than no duct	S/E	Chew 2021 p. 14; Dayhoum 2025 ApplSci p. 9; Martin & Tung 2004 pp. 3–4; Zawodny 2025 p. 9
2. Inlet lip: generously rounded	16 cm rotor, 17 shrouds: more lip radius always helped; best model 13 % D_t. Plain cylinder, lip 1.4→8.3 % D_t: figure of merit 0.72→0.89. Recommended 3–15 % of rotor diameter. Below ~6 % D_t the lip separates (one study still separated at 12.5 %). A smaller lip lowers duct thrust but steadies hover in a crosswind.	minimum ~12 mm; aim 20–26 mm radius	E/R	Pereira 2008 pp. 1, 70, 113; Dayhoum 2025 p. 9; Zhang & Barakos 2020 p. 23; Martin & Tung pp. 3–4
3. Wall at least 10 % of D thick at the propeller plane	Black & Wainauski: thinner walls separate at static conditions. It is also the room the rounded lip needs.	≥ 20 mm	R	Zawodny 2025 (NASA TM) p. 8
4. Propeller set back behind the lip	NASA rig: propeller 45 % of the duct chord behind the leading edge. Hamilton Standard: the rearward position is better static, the forward one in forward flight.	disc clear of the lip curve; ~45 % of duct length in	E/R	Zawodny 2025 p. 9; Pereira 2008 p. 78
5. Outlet: a modest flare (second-order)	Best diffuser angle 10°, length 31–50 % D_t. Exit area ratio called “the most powerful” variable: up to +15 % static thrust at 1.2–1.3; NASA chose 1.15 (exit area / disc area). Sources disagree on the angle (see §04); diffuser length moved thrust only ~3 % in one sweep.	10°, 63–102 mm long; exit ≈ 218 mm dia. at 1.15	E/R	Pereira 2008 pp. 1, 75, 77, 113; Zawodny 2025 p. 8; Dayhoum 2025 p. 9
6. Supports out of the rotor plane	Fenestron tests: moving struts ~5.5 % D_t further from the rotor cut noise by up to 10 dB, to the no-strut level. Acoustic, not thrust.	struts ≥ ~11 mm behind the disc	R	Pereira 2008 p. 119

The passages the first two rules stand on

> “While the tip clearances of 0.6% and 1.1% produce comparable results, a sudden drop in performance can be observed for 3.6% tip clearance.”
>
> Chew, Gan & Hesse, AIAA SciTech 2021, p. 14

> “increasing the tip clearance from 3.04% to 5.17% of the rotor radius can lead to a significant reduction in hover efficiency of up to 18.1%, even falling below the performance of an open rotor configuration.”
>
> Dayhoum, Ramirez-Serrano & Martinuzzi, Appl. Sci. 15 (2025) 9594, p. 9

> “As the tip gap is increased, the suction pressure near the leading edge rapidly collapses”
>
> Martin & Tung, AHS 60th Forum 2004, p. 3

> “Increasing the lip radius and decreasing the blade tip clearance caused performance to improve, while optimal values of diffuser angle and length were found to be 10° and 50% of the shroud throat diameter, respectively.”
>
> Pereira, PhD dissertation, Univ. of Maryland 2008, p. 1

> “The increase in rlip caused a near-linear increase in figure of merit (at the same collective) from 0.72 to 0.89.”
>
> Pereira 2008, p. 70, reporting Parlett 1955

> “it is argued that there could be a critical lip radius of 6% duct diameter, below which separation would take place. However, as tested by Graf et al. [31], flow separation was also observed with a radius of 12.5%.”
>
> Zhang & Barakos, Aeronautical Journal 124 (2020), p. 23

03 What a good duct can give — and why a small one can lose

> “Compared to the baseline open rotor, the shrouded rotors showed increases in thrust by up to 94%, at the same power consumption, or reductions in power by up to 62% at the same thrust.”
>
> Pereira 2008, p. 1 — 16 cm rotor, best of 17 shrouds

> “the duct thrust developed to approximately 38% of the isolated rotor thriist [sic] at high RPM.”
>
> Martin & Tung 2004, p. 3 — 10-inch rotor, smallest tip gap (1 % R)

> “The internal flow losses tend to counteract the duct thrust, and beiow [sic] a certain RPM the duct produces a net download (compared to the isolated rotor).”
>
> Martin & Tung 2004, p. 3

> “for a given overall aircraft size limitation, and ignoring considerations of the safety benefits of a shroud, a larger-diameter open rotor is more likely to give better performance than a smaller-diameter shrouded rotor.”
>
> Pereira 2008, p. 2

So the duct's thrust is suction on the inlet lip: a wide gap or a sharp lip kills it, and at small scale the viscous losses inside the duct can then outweigh what is left. Our unlined duct (§01) sits on the losing side. The gain is not scale-free either: over a similar tip-clearance range Hubbard's 48-inch model lost ~84 % of its thrust and Mort's 7-foot fan ~20 % (Pereira 2008, p. 70, 74), so no percentage here transfers to an 8-inch duct without measuring it.

04 Where the sources disagree

- **Diffuser angle.** Pereira finds 10° best; Taylor 1958 finds the result “virtually independent of the expansion angle of the diffuser, and depended only on the value of the expansion ratio” (Pereira p. 74); Dyer found “increasing the diffuser angle caused thrust and power loading to decrease, contrary to predictions” (Pereira p. 94); Li, Yonezawa & Liu's CFD put the optimum at 0° (Drones 2021, p. 10). Keep the flare modest and treat it as a later experiment.
- **Diffuser length.** Pereira gives a single optimum (31–50 % D_t); Ma’arof 2023 found no correlation with thrust and named tip clearance the most consistent driver (p. 314).
- **How small the gap must be.** On 30-inch propellers a gap of 0.12→0.56 % D_t cost under 5 % thrust (Pereira p. 79), while 0→4.4 % D_t cost Hubbard ~84 % (p. 70): the penalty grows once the gap is already a few percent, which is where a 3–5 mm gap on our duct would be.
- **Lip radius against crosswind.** The larger lip gives more static thrust; Martin & Tung's smaller lip “improved the stability for hover in a crosswind” (p. 4). On the stand only the first matters.

05 How to test it on our rig

1. **Record the duct first:** inner diameter, tip gap, duct length, where the propeller disc sits along it, and the present inlet edge (sharp, chamfered or rounded). Without these the millimetres in §02 are guesses.
2. **One change at a time, against same-day references.** Each session: bare propeller and unlined duct at the start (and the bare propeller again at the end), then the variant; repeats of one set-up agree within 0.1 N, so differences above ~0.2 N are real.
3. **Order:** (a) a bolt-on rounded lip ring, ~25 mm radius, on the existing duct; (b) a tighter tip gap (the linings already show the direction); (c) only then the outlet flare.
4. **Compare at equal rpm or per watt**, not only at equal ESC signal: the bare propeller ran 4 % slower than the ducted one, and the NASA rig needed 4.6 % more rpm with the duct to match the open propeller's thrust (Zawodny 2025, p. 49).
5. **Photograph every set-up.**

Derived 2026-11-02 from `papers/SoundVisualizer/design-rules/EXTRACTS-DUCT-LINING.md` §1–1.4 (private repo `asdfgh0318/papers`, commit 4c47a05), which governs; every quotation on this page was re-checked against the `pdftotext` dumps with the alphanumeric-key matcher of `scripts/verify_quotes.py`. Sources read at source: Pereira 2008 (PhD, Univ. of Maryland); Martin & Tung 2004 (AHS Forum 60, OCR text — misspellings kept and marked [sic]); Chew, Gan & Hesse 2021 (AIAA SciTech, doi:10.2514/6.2021-0212); Dayhoum, Ramirez-Serrano & Martinuzzi 2025 (Appl. Sci. 15, 9594); Zawodny, Schiller, Pettingill & Medina 2025 (NASA/TM-20250009190); Zhang & Barakos 2020 (Aeronaut. J. 124, doi:10.1017/aer.2019.164); Li, Yonezawa & Liu 2021 (Drones 5, 101); Ma’arof et al. 2023 (JARASET 32(3)). Parlett, Hubbard, Taylor, Mort, Black/Wainauski/Rohrbach and Dyer are reported inside Pereira 2008 and were not retrieved themselves (evidence R). Our measurements in §01 are captures on the Pi, backed up to `SoundVisualizer-data`.